

5-Mark Questions & Answers MPCA Module 6: Pentium 4 & ARM Processor

Q1. Explain Pentium 4 NetBurst microarchitecture, Trace Cache, and Rapid Execution Engine.

- 20-stage Hyper-Pipeline.

- 12h

Q2. Explain Hyper-Threading Technology (HT) architecture, benefits, and hardware implementation.

- Duplicated state (GPRs, EFLAGS, CR0-CR3) vs shared execution units (ALUs, FPU, Caches).

- 15-

Q3. Explain ARM RISC Processor architecture, Load-Store design, Barrel Shifter, and CPSR/SPSR registers.

- Strict Load-Store architecture (LDR/STR).

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